

Sai Ravi Teja Gangavarapu

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EDUCATION

Carnegie Mellon University

Master of Software Engineering – Scalable Systems (ML Systems, Statistics, QA, Formal Methods)

Pittsburgh, PA

Aug 2025 – Dec 2026

Mahindra École Centrale

Bachelor of Technology in Computer Science

Hyderabad, India

Aug 2020 – Dec 2023

University of Florida

Senior Cert + Master's coursework in Computer Science (Algorithms, UX, Graphics – 4.0)

Gainesville, FL

Jan 2024 – May 2024

SKILLS

Languages: Python, C++, Go, TypeScript, SQL, x86 Assembly

ML / AI: PyTorch, CUDA, GPU kernels, distributed training; 2D/3D perception (segmentation, detection, LiDAR/RGB-D, sensor fusion); generative modeling (diffusion, flow matching); model optimization & deployment (TensorRT, ONNX, INT8 quantization); LLM fine-tuning (LoRA/PEFT), tool calling, RAG, LangGraph

Systems & Infrastructure: AWS (Athena, S3, EC2, Lambda), Docker, Kubernetes, CI/CD (ArgoCD, GitHub Actions), PostgreSQL, Redis, microservices, large-scale data pipelines (Polars/Pandas), OpenMP, SIMD, HPC

Robotics: ROS2/Nav2, simulation & sim-to-real, motion planning & control (MPPI/MPC, Hybrid A*), state estimation

Web: React, React Native, FastAPI, GraphQL, REST APIs, WebSockets, Next.js

Relevant Coursework: ML Systems (15-642), Distributed Systems, Software Architecture, QA & Testing, Visual Learning & Recognition (16-824), Diffusion & Flow Matching (10-799), Planning & Decision-making in Robotics (16-782), F1TENTH Autonomous Racing (audited)

EXPERIENCE

Researcher – AirLab, CMU Robotics Institute (Adversarial Robustness & V&V)

Aug 2026 – Present

- Researching adversarial robustness of learning-based drone perception and obstacle-avoidance algorithms under Prof. Sebastian Scherer; building an agentic, automated test bench in Isaac Sim to systematically discover and characterize failure modes across nominal, off-nominal, and adversarial scenarios.

Autonomy ML Systems Intern – Motional

May 2026 – Aug 2026

- Built kinematics-aware, multimodal evaluation metrics for trajectory prediction, surfacing regressions missed by prior methods, and a production dashboard to deep-dive into regressions across maneuver types and model versions on a Level-4 self-driving stack covering millions of agents.
- Engineered the underlying data platform in SQL, AWS Athena, and Python (Polars/Pandas) to process ~1 PB of large-scale driving logs and surface prediction quality and calibration across the fleet.

Autonomy Subsystem Lead – CMU Lunabotics (NASA Autonomy Award)

Jan 2026 – Present

- CMU Lunabotics is a NASA competition to build a lunar robot that autonomously digs and dumps regolith while evading rocks and craters. Leading the autonomy sub-team, advised by Prof. Wenshan Wang. Implemented the rover's ROS2 navigation stack with GPU-optimized MPPI control; tuned through lunar-simulant field testing.
- Built a Unity simulation of the competition lunar arena to generate synthetic training data; trained an instance segmentation model (PyTorch/YOLOv8) for onboard obstacle detection and quantized it FP32 → INT8 via TensorRT for the rover's Jetson Orin – cutting model size 65.6% with only ~3% mIoU drop.
- Built a LiDAR + camera sensor-fusion pipeline for 3D obstacle detection using confidence-weighted range fusion, validated against simulated ground truth (IoU, precision, recall).

Founding Software Engineer – Tapsta

Jul 2024 – Jun 2025

- Architected end-to-end full-stack social rewards mobile application using React, React Native, FastAPI, and PostgreSQL, serving 1,500+ users across iOS and Android.
- Designed database architecture with 25+ schemas and optimized client-side operations with caching and debouncing, achieving 30% faster response times. Worked with product team to redesign onboarding flow, driving 40% user retention.
- Orchestrated Plaid API integration alongside REST API creation, establishing secure bank connectivity and enabling automated cashback processing and ACH transfers for all application users.
- Built a containerized CI/CD pipeline (Docker, Kubernetes, AWS, ArgoCD) cutting deploy time 70%; led a team of 3 SDE interns, establishing code review processes contributing to 30% faster feature delivery.
- Built a semantic people search engine for alumni networks using embedding-based similarity (LangGraph).

Research Assistant – Fan Lab, University of Florida

Feb 2024 – Jun 2024

- Applied genomic models for rare disease prediction using transformer architectures, processing 15,000+ DNA sequences; developed MLP models using protein embeddings (ESM3) analyzing RNA-binding proteins.

SDE Intern – Hashira.io

Apr 2023 – Dec 2023

- Fine-tuned LLMs (LLaMA/BERT) for automated tool-calling on production payroll systems, building function-calling workflows reaching 90% natural language to API accuracy and reducing API integration time by 25%.
- Built a full-stack crypto analytics platform (React, FastAPI, Go, PostgreSQL) with real-time monitoring dashboard and a leaderboard microservice for garden.finance, supporting \$150M+ trading volume; improved backend performance with Go and PostgreSQL optimizations for 20% faster queries.

Research Assistant – Mahindra École Centrale

Dec 2022 – Jan 2025

- Published research on emotion-targeted music generation using FFT, differential evolution, Fuzzy C-means, self-organizing maps achieving 85% accuracy in classification (IEEE CEC 24). Implemented ALI-GAN model with t-SNE and PCA for clustering analysis on 1000+ samples. Built MIR pipelines extracting 100+ features.

PROJECTS

Blackwell GEMM Kernel & Operator Fusion | TVM/TIR, C++, CUDA | Prof. Tianqi Chen, ML Systems (15-642)

- Built a warp-specialized FP16 GEMM kernel for NVIDIA Blackwell (SM100) with TMA async loads, tensor-core MMA, multi-stage pipelining, and persistent-kernel tile scheduling; added autotuned operator fusion to compile and deploy optimized kernels.
- Reached 1,400–1,600 TFLOPs on saturated matrix dimensions (75–85% of cuBLAS throughput), validated for correctness against a cuBLAS reference.

SAPS: Step-Aware Pruning Schedule for Diffusion LLMs | PyTorch, LLaDA-8B, ML Systems (15-642)

- Replaced the fixed KV-cache pruning ratio in Sparse-dLLM with an exponentially decaying, step-aware schedule (keep 70% early, 10% late), requiring no retraining and adding one floating-point op per denoising step.
- Cut KV cache from 145 MB to 50 MB (66% less memory) on LLaDA-8B/GSM8K while matching vanilla accuracy (78.2% vs 78.17%); a Pareto improvement over Sparse-dLLM (31.4% less memory, +1.9pp accuracy). Ran layer-aware and pruning-floor ablations.

Edge-Guided Flow Matching for Controllable Face Synthesis | Diffusion & Flow Matching (10-799) | **GOAT Award (Best Poster)**

- Built EdgeAdapter, injecting edge features into UNet skip connections and the middle block rather than concatenating at the input, with cycle loss and timestep-weighted conditioning; improved every controllability metric and halved perceptual distance (LPIPS 0.34 → 0.17), with the realism tradeoff characterized via ablations.

CMR Driverless (Carnegie Mellon Racing) | ROS2, LiDAR, Camera, Motion Planning

- Perception engineer on CMR's driverless Formula SAE car; contributing to the LiDAR-camera cone detection and classification pipeline in ROS2.

ML-Based Turbine-Rotor Defect Detection Platform | CMU MSE Practicum · Siemens Energy | Python, PyTorch, Signal Processing

- Building an end-to-end platform that ingests hundreds of phased-array ultrasonic (UT) scan files per inspection and runs ML anomaly detection and classification to flag turbine-rotor defects for engineer review; owning the data pipelines, model training/validation, and reporting UI.

gitask.org | WebGPU, TypeScript, React | **200+ users**

- Offline, in-browser RAG tool to chat with any GitHub repo via URL swap (github.com → gitask.org). Local LLM inference via WebGPU. AST chunking, graph-aware hybrid search, binary quantized embeddings – zero data leaves the browser.

GPU & CPU Performance Engineering | CUDA, C/C++, OpenMP, x86 Assembly, SIMD

- Built a GPU-accelerated ray tracer in CUDA achieving 1,600x speedup over CPU with real-time FFT-based audio reactive 256-agent boids simulation; separately built CPU-optimized parallel algorithms using OpenMP with 10x speedup and x86 assembly with SIMD optimizations. Also built modular 3D renderer in C++ and OpenGL.

F1TENTH Autonomous Racing (audited) | ROS2, Simulation, Motion Planning

- Completed CMU's F1TENTH course in simulation: processed raw LiDAR scans for reactive obstacle avoidance (Follow the Gap) and wall following, plus path tracking on 1/10-scale vehicle dynamics.

RootCause Auto-Remediation Agent | Next.js, OpenTelemetry, MongoDB, FastAPI, Gemini

- Built a self-healing agent that parses distributed traces via OpenTelemetry, deduplicates alerts, and auto-generates syntax-validated (AST) code fixes via Gemini, cutting mean time to resolution.

RainPod: AI-Powered Interactive Podcast | CrewAI, Python, FastAPI, React | **300+ users**

- Built a multi-agent system (CrewAI) that converts PDFs and topics into interactive AI podcasts – agents handle research, script generation, and audio synthesis in an orchestrated pipeline.

Project RECON | Raspberry Pi, OpenMPI, GlusterFS | \$2,000 funding

- Architected 8-node Raspberry Pi 4B compute cluster with distributed storage (GlusterFS), LDAP authentication, and Slurm job scheduling, serving 400+ students for coursework.

OneAIClick.com | Python, React.js, HuggingFace, PEFT/LoRA

- Developed an LLM fine-tuning abstraction tool for HuggingFace models that cut development cycles by 60% and reduced boilerplate code by 75%, enabling rapid idea validation with full data privacy.

Interpretable & Controllable Text-to-Audio Synthesis | PyTorch, CLAP, TorchSynth, Sound Design

- Built a text-conditioned synthesizer controller maximizing text-audio embedding similarity (CLAP). Also did inverse synthesis.

SimplySecure – macOS Security Scanner | SwiftUI, SwiftAIAgents, Gemini, Retell AI

- Built a gamified macOS security scanner with AI phishing call simulation, agentic privacy policy analysis for threat training.

LEADERSHIP AND ACHIEVEMENTS

6x hackathon winner:

2nd Place, TartanHacks 2026 (1,000 participants, \$3K prize from AppLovin).

1st Place, WaffleHacks 2024 (320+ participants): Built FlashFocus – a Chrome extension to block distractions and turn them into learning with AI flashcard quizzes using RAG. (React, FastAPI, SQL, Groq)

1st Place, Talentmapp23 (300+ participants): Intelligent task-tracking app with LLM suggestions (React, GPT, Flask).

1st Place, NR Hackathon 2023 (50 teams): Engineered a distributed cluster implementing the Raft consensus protocol from scratch.

3rd Place, CMU AWAP 2026 (~80 teams): Multi-agent bot (no RL) – Hungarian algorithm for task assignment, Johnson's algorithm + BFS for routing.

Buildspace S5 30+ users – Desktop app, AI sample packs from songs with stem separation, drum extraction & flips for musicians.

President, Enigma, the Computer Science Club (2021-2023): Conducted technical workshops on Gamedev, ML, Linux, heading outreach initiatives increasing the club membership by 40%, reaching over 2000 students. Collabs – Ubisoft, NVIDIA.

PUBLICATIONS

Emotion Aligned Music Composition from Sound Fundamentals Using Differential Evolution – IEEE Congress on Evolutionary Computation (CEC 2024)

From Clicks to Insights: Analysing Online Customer Reviews for Handicraft Products – Technology Analysis & Strategic Management (2025)